WEEK 8 · MODULE 4 — AGENTS

Agent Failure Modes & Trajectory Evals

Build week

Spot how your agent fails, defend it against tricks, and measure a fix to its worst problem.

What this week is about

Agents fail in new and sneaky ways — going in circles, picking the wrong tool, or reaching a right answer through a lucky wrong path that will break later. This week you learn to watch the whole path, not just the final answer, and to protect the agent from being tricked.

Why it matters

If the answer is right, why care how it got there?

A right answer reached by luck won't stay right. Next week the same lucky path gives a wrong answer, in production, to a customer.

What is "prompt injection"?

When hidden instructions inside a document or web page trick your agent into doing something it shouldn't. The agent can't tell your instructions from the text it reads.

Why give tools the least access possible?

Your agent is helpful and holds your keys. If something goes wrong, tightly-scoped tools keep the damage small.

What you'll learn

- The common agent failures: loops, wrong tool, made-up inputs, giving up quietly
- Judging the whole path, not just the final answer
- The gap between "got the right answer" and "took the right steps" — and why it matters
- Prompt injection: hidden instructions in documents or web pages that hijack your agent
- Defending the agent: checking what it reads, and limiting what each tool can do
- Fixing the worst failure and measuring that it actually went down

Topics covered

The exact concepts to study:

- Agent failure modes · Trajectory evaluation
- Expected tool sequences · Tool-choice accuracy
- Outcome vs trajectory gap · Cost per task (mean & p99)
- Prompt injection (direct) · Indirect prompt injection
- Tool sandboxing & least privilege · Output validation
- OWASP LLM Top 10

Your task this week

Evaluated	Week 9 · Monday

Look at a batch of your agent's runs and find where it took a wrong path even when the answer looked right. Try a simple attack that hides an instruction in a document, then defend against it, and fix your top failure — measuring the improvement.

Choose your track

Same task, six different topics — so everyone works on their own version. Download whichever you're assigned.

	Topic	Task
A	**Customer support tickets**	Find the outcome-vs-trajectory gap in the ticket agent, then close one mode
B	**Recipes & food**	Find the outcome-vs-trajectory gap in the recipe agent, then close one mode
C	**HR policy**	Find the outcome-vs-trajectory gap in the HR agent, then close one mode
D	**Insurance claims**	Find the outcome-vs-trajectory gap in the claims agent, then close one mode
E	**Developer documentation**	Find the outcome-vs-trajectory gap in the docs agent, then close one mode
F	**Legal contracts**	Find the outcome-vs-trajectory gap in the contract agent, then close one mode

How your mentor checks

When	Friday · no marks

A quick review of your task. Not a test — just a few things your mentor looks at, so you can fix anything before Monday.

- Did they find a case where the answer was right but the path was wrong?
- Did they successfully trick their own agent, then stop the trick?
- Is there a before-and-after number on their top failure?
- Can they name what could still get through?